American Computer Science League

2023-2024 ● Contest 1: ACSL Tiles ● Junior Division

PROBLEM: ACSL Tiles is a one-person game played with rectangular tiles. Each tile has a single-digit number between 1 and 9 inclusive at each end. At the start of the game, there are 4 rows, each with a number. The goal of the game is to build rows by placing a tile at the right end of the topmost row whose last number matches a number on the tile. Tiles can be re-oriented; thus, the tiles 34 and 43 are the same tile. If a tile cannot be placed on any row, it is placed in the *discard pile*. When all tiles have been played or discarded, find the sum of the single-digit numbers on all of the tiles in the *discard pile*.

EXAMPLE:

Input:

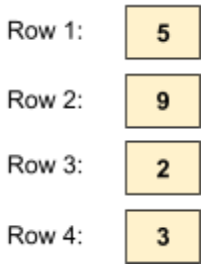
```
5923
56 85 27 73 14 34 62
```

Output:

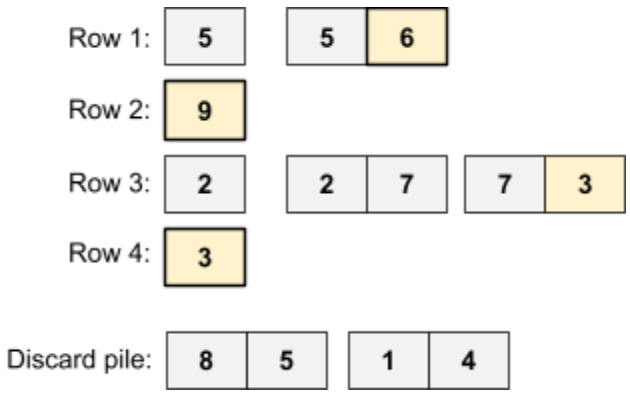
```
18
```

Explanation:

The game starts with 4 rows having numbers 5, 9, 2, 3.



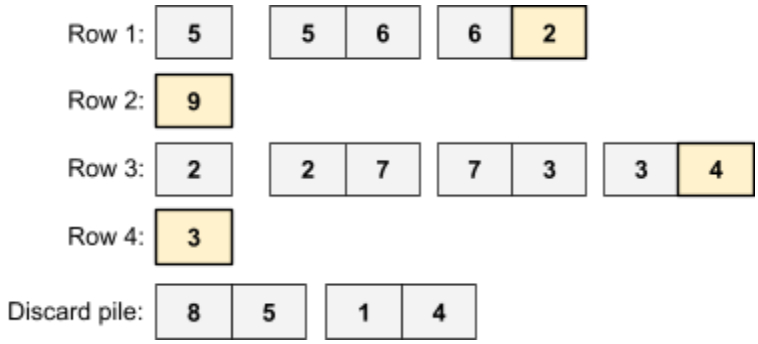
After tile 56 is placed in Row 1, tile 85 is placed in the discard pile. Both 27 and 73 are placed in Row 3 and tile 14 is placed in the discard pile.



The next tile, 34, is placed in Row 3, the topmost row it can go on. And finally, the 26 tile is added to Row 1, with the orientation of 62. The result is as follows:

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The result is the sum of 8+5+1+4=18.

INPUT: Input a 4-digit integer that gives the initial numbers from Row 1 to Row 4, and a string of no more than fifty 2-digit integers, each separated by a single space. Each 2-digit integer represents the two numbers on each tile, both between 1 and 9, inclusive.

OUTPUT: After placing the tiles using the rules above, output the sum of the single-digit numbers on all of the tiles in the discard pile.

SAMPLE INPUT:

```
1. 5923
   56 85 27 73 14 34 62
2. 8423
   74 92 57 93 26 87 14 63 82 54 12
3. 1253
   51 81 35 84 95 26 59 13 71 35 46 28
4. 2694
   69 76 41 89 16 78 64 36 12 95 52
5. 6479
   58 73 92 54 75 35 78 25 81 24 16 95 36 82 14 27 43 13 51
```

SAMPLE OUTPUT:

```
1. 18
2. 26
3. 31
4. 22
5. 45
```

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TEST DATA

TEST INPUT:

1. 3972
   18 17 65 61 37 51 57 38 72 92 54 59 43 41 31 28
2. 9146
   95 74 51 19 75 26 32 39 35 31 25 73
3. 7918
   63 18 56 98 48 52 26 92 83 13 17 58 91 67 58
4. 9758
   52 14 51 27 77 62 76 82 96 56 46 49 87
5. 7169
   71 56 15 65 98 71 89 71 11 55 77 17 66 51

TEST OUTPUT:

1. 56
2. 0
3. 59
4. 48
5. 14

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问题: ACSL Tiles 是一个与长方形图块有关的单人游戏。每个图块的两端都有一个位于 1 ~ 9 之间（包括 1 和 9）的个位数。游戏开始时，共 4 行，每一行都包含一个数字。游戏目标是构建行，将一个图块放在最顶层一行的右端，使得这行的最后一个数字与图块上的一个数字匹配。图块可以改变方向；例如，图块 34 和图块 43 相同。如果一个图块不能放置在任何一行中，则放弃该图块并将其归入 discard pile。当所有图块都已放置或放弃时，求 discard pile 中所有图块上个位数的总和。

示例:

输入:

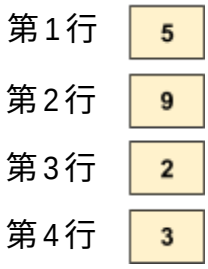
```
5923
56 85 27 73 14 34 62
```

输出:

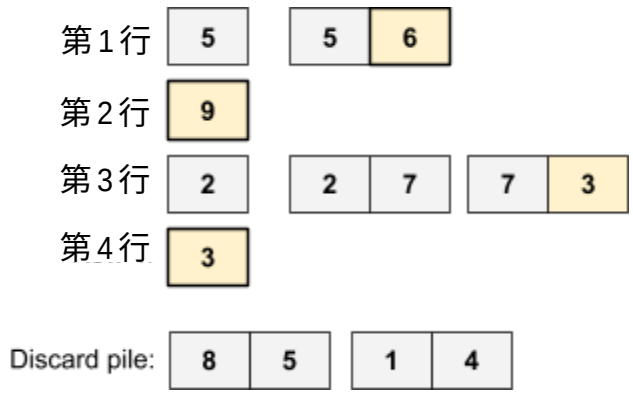
```
18
```

详解:

游戏开始时，4 行初始数字为 5, 9, 2, 3。



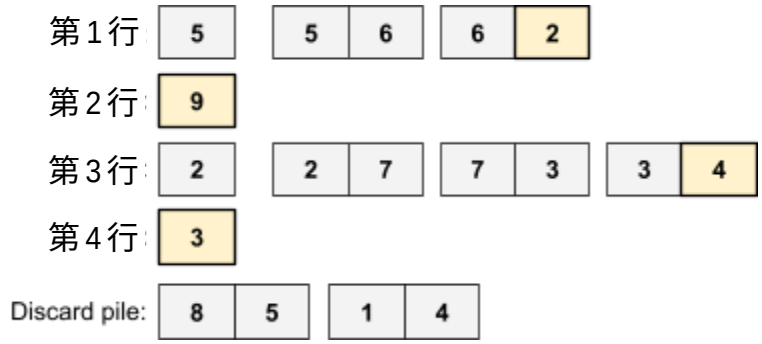
首先将图块 56 放在第 1 行，然后图块 85 归入 discard pile 。图块 27 和 73 都放在第 3 行，图块14 归入 discard pile。



接下来，将图块 34 放在第 3 行，而后返回第 1 行。最后，图块 26 旋转变成 62 后放在第 1 行。结果如下：

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加总可得： 8+5+1+4=18。

输入: 输入一个 4 位数，这 4 位数字为第 1 行到第 4 行的初始数字, 后面跟着一串包含不超过 50 个的两位整数，数字之间用一个空格隔开。每个两位整数代表每个图块上两个位于1 - 9 之间（包括 1 和 9 ）的数字。

输出: 按照上述规则放置图块之后，输出 discard pile 中所有图块上个位数的总和。

样例输入：

```
1. 5923
   56 85 27 73 14 34 62
2. 8423
   74 92 57 93 26 87 14 63 82 54 12
3. 1253
   51 81 35 84 95 26 59 13 71 35 46 28
4. 2694
   69 76 41 89 16 78 64 36 12 95 52
5. 6479
   58 73 92 54 75 35 78 25 81 24 16 95 36 82 14 27 43 13 51
```

样例输出:

```
1. 18
2. 26
3. 31
4. 22
5. 45
```

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测试数据

测试输入:

1. 3972
   18 17 65 61 37 51 57 38 72 92 54 59 43 41 31 28
2. 9146
   95 74 51 19 75 26 32 39 35 31 25 73
3. 7918
   63 18 56 98 48 52 26 92 83 13 17 58 91 67 58
4. 9758
   52 14 51 27 77 62 76 82 96 56 46 49 87
5. 7169
   71 56 15 65 98 71 89 71 11 55 77 17 66 51

测试输出:

1. 56
2. 0
3. 59
4. 48
5. 14